

CADASTRAL DATA ACQUISITION TECHNOLOGIES ASSIGNMENT

Land rights regularization: a Fit-For-Purpose (FFP) approach

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1. INTRODUCTION

This report outlines an affordable way to update Enschede Municipality's land records, focusing on both the cadastral and register parts. The cadastral map hasn't been updated since 1967, leading to parcels being split up without proper registration. The main aims here are to bring the map up to speed, register current land rights, input all the info into a database that meets LADM standards, allow public access to the new parcel boundaries, and issue fresh land titles.

To make this happen, we used data from a participatory mapping exercise and a list of existing land rights. The report includes a diagram that meets LADM standards, details on the tools and accuracies used for fieldwork, sample land titles for the new parcels, and a discussion on the system's pros and cons. plus, there are appendixes with GIS datasets and projects, as well as URLs for services and a web map application. All the necessary details are covered!

2. Documenting the new data structure

Enschede Municipality has requested that data be entered into the database following the Land Administration Domain Model (LADM) guidelines. To meet this requirement, we've crafted LADM-compliant class diagrams using Enterprise Architect version 15.2 software. The entities, including party, RRR (Right, Responsibility, Restriction), and Spatial Unit tables, are designed in line with LADM standards. Attributes in the diagrams adhere to Dutch laws. To ensure data consistency, code lists have been integrated into the LADM model. Additionally, we've employed an area calculation function within the Spatial Unit to assess the quality of spatial data. Below, you'll find the LADM model created with the Enterprise Architect program.

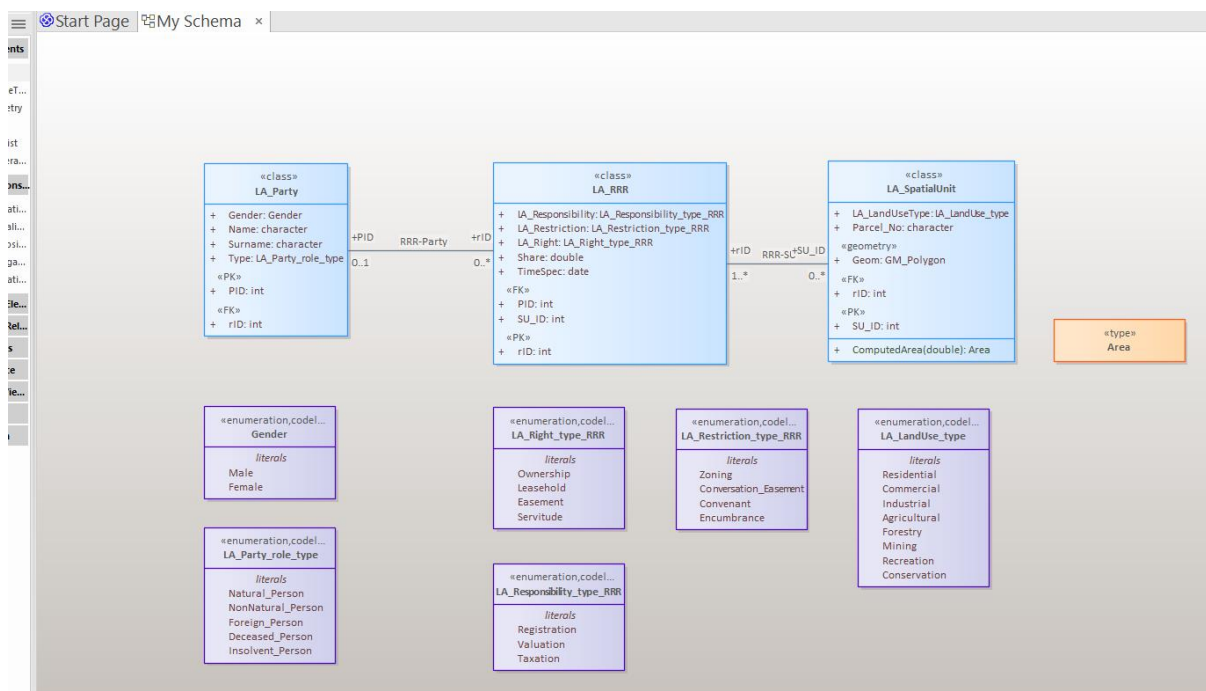


Figure 1 LADM from enterprise

3. Conduct the fieldwork

Measurements of parcels lacking cadastral registration were conducted using QField program version 2.7.3. We opted for QField to save time and money as its open-source software under the GNU General Public License version 2.0. However, since QField relies on mobile device GPS accuracy, our measurements had an accuracy of approximately 6 meters. To utilize QField, we first created layers of tables and attributes in the QGIS program on a computer. We established a "new parcel" layer at this stage, which should help in populating LADM tables. Within the QGIS-created table, code list values were implemented for gender, party role type, and responsibility, right, and landusetype columns. Once the layer was saved in qgs format, it was transferred to the QField program for field measurements.

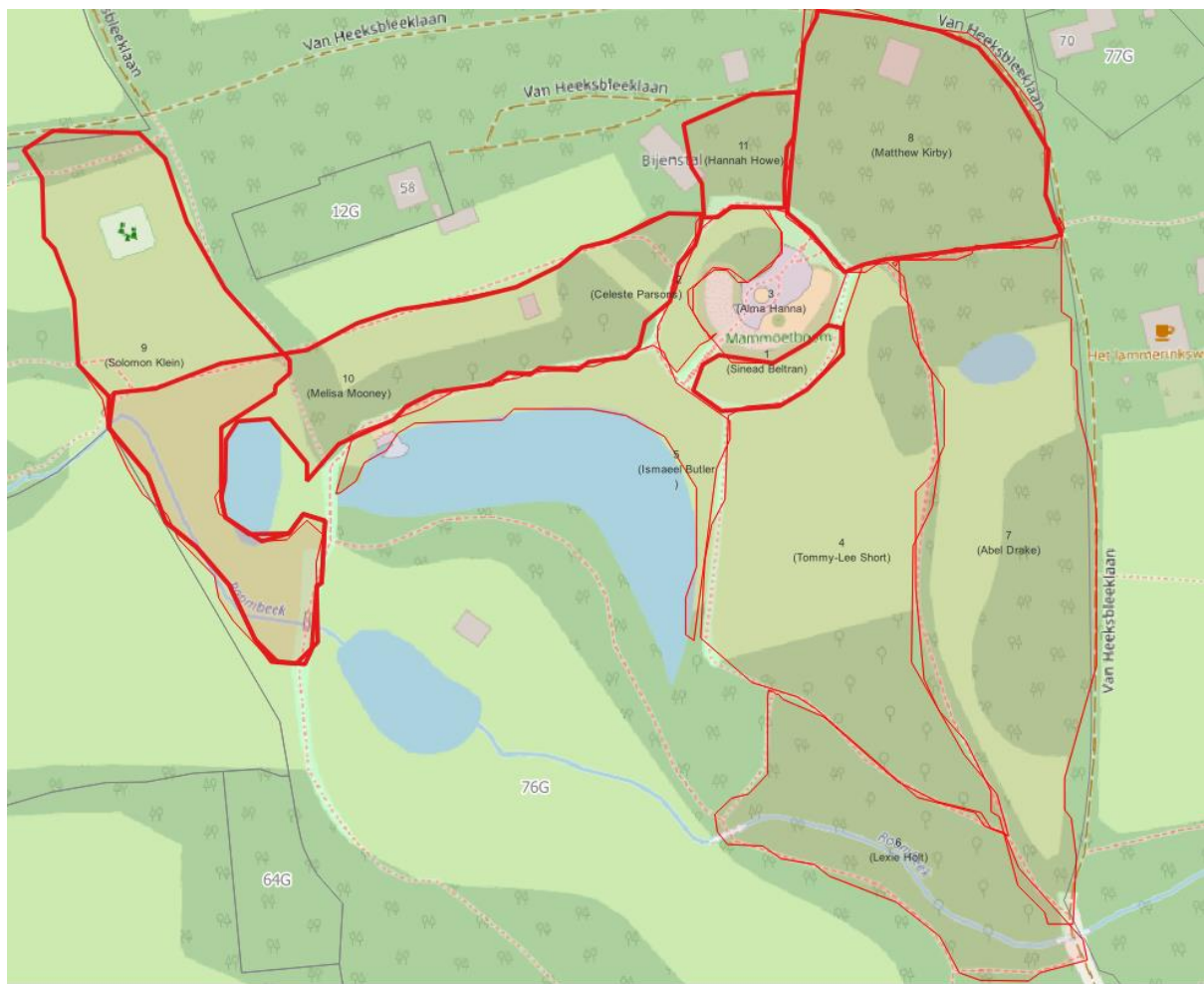


Figure 2 QField data collection

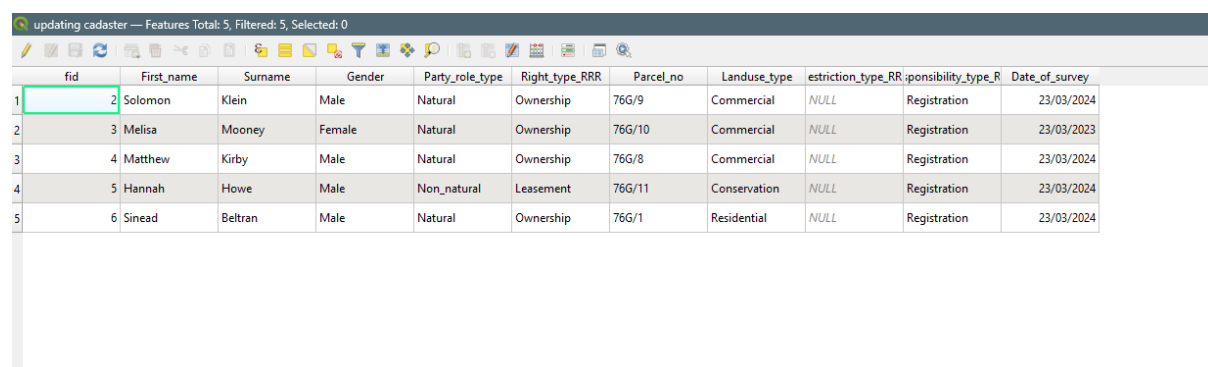
4. Implementing the data base

To meet the demand for a database approach in the community of Enschede, we used GEO PACKAGE which is the core element. To have historical land records intact ever since 1967 and a deficient way of registering parcels, data consolidation had been core. The GeoPackage became a backbone of digital transformation as the format that unifies the land data in the city was provided.

The initial activity was to define the GeoPackage and create a digital repository for registered Enschede's land. It was the start of a totally new kind of process with electronic record-keeping not only just a model but actually something real for the municipality. In the GeoPackage, all the tables and attributes were pre-configured to the destination according to the digital terrain types. Boundaries, names, and other details derived, were designated and arranged for easy access and manageability. Due to the particularity of each field and data type, the whole set of database was built in such a way that its integration and coherence could be assured.

Real time transactions exchanged data in to the database from different sources and the plots were assigned and document details were recorded in a smoothening way into the GeoPackage package. It was universal in nature and inclusive of the both spatial and non-spatial data to the view of the entire municipality's cadastre.

The GeoPackage had more potency in the operations level rather than just being a simple storage system. It, therefore, provided the opportunity to understand data in the manner it was intended and conveyed the insights. Though QGIS was one of the tools to be used, it revealed by spatial data the information which had been buried in the deep. The database became more and more sophisticated and eventually simpler and easier to use by the end-user. The GIS software and libraries used in the Project gave the stakeholders the privileged of examining the data held within the GeoPackage, enabling the decision-making operation to be more informatively based.



	fid	First_name	Surname	Gender	Party_role_type	Right_type_RRR	Parcel_no	Landuse_type	Restriction_type_RR	Responsibility_type_R	Date_of_survey
1	2	Solomon	Klein	Male	Natural	Ownership	76G/9	Commercial	NULL	Registration	23/03/2024
2	3	Melisa	Mooney	Female	Natural	Ownership	76G/10	Commercial	NULL	Registration	23/03/2023
3	4	Matthew	Kirby	Male	Natural	Ownership	76G/8	Commercial	NULL	Registration	23/03/2024
4	5	Hannah	Howe	Male	Non_natural	Leasement	76G/11	Conservation	NULL	Registration	23/03/2024
5	6	Sinead	Beltran	Male	Natural	Ownership	76G/1	Residential	NULL	Registration	23/03/2024

Figure 3 Geo Package Database

5. Dissemination

So far, the focus has been on measuring new parcels in the field and recording them geo package in a format compliant with the Land Administration Domain Model (LADM). Now, attention shifts to disseminating the acquired data. QGIS version 3.22.9 - Białowieża has been selected for this purpose, just as it was for data collection.

For dissemination, the qgis2web plugin, a feature of the QGIS software package, has been utilized alongside QGIS CLOUD to provide a geo-web service. Leaflet version 1.7.1 has been chosen for the web-map application. The published data consists of the updated cadastre parcels layer measured using QField, available in both HTML and WMS formats.

The below figure, presents a screenshot of the HTML web page. When a user hovers over a parcel on the web page, data from the attributes table of the new parcels layer is displayed. The link to the web page is provided in the Appendix section.

As the data is in HTML format, the link containing the data will be used for display purposes, leveraging its various advantageous features that enhance the appeal of the maps. Additionally, a WMS link is generated through the use of the QGIS Cloud plugin, facilitating easy sharing of the new parcels layer as a WMS link that can be readily utilized within the QGIS program.

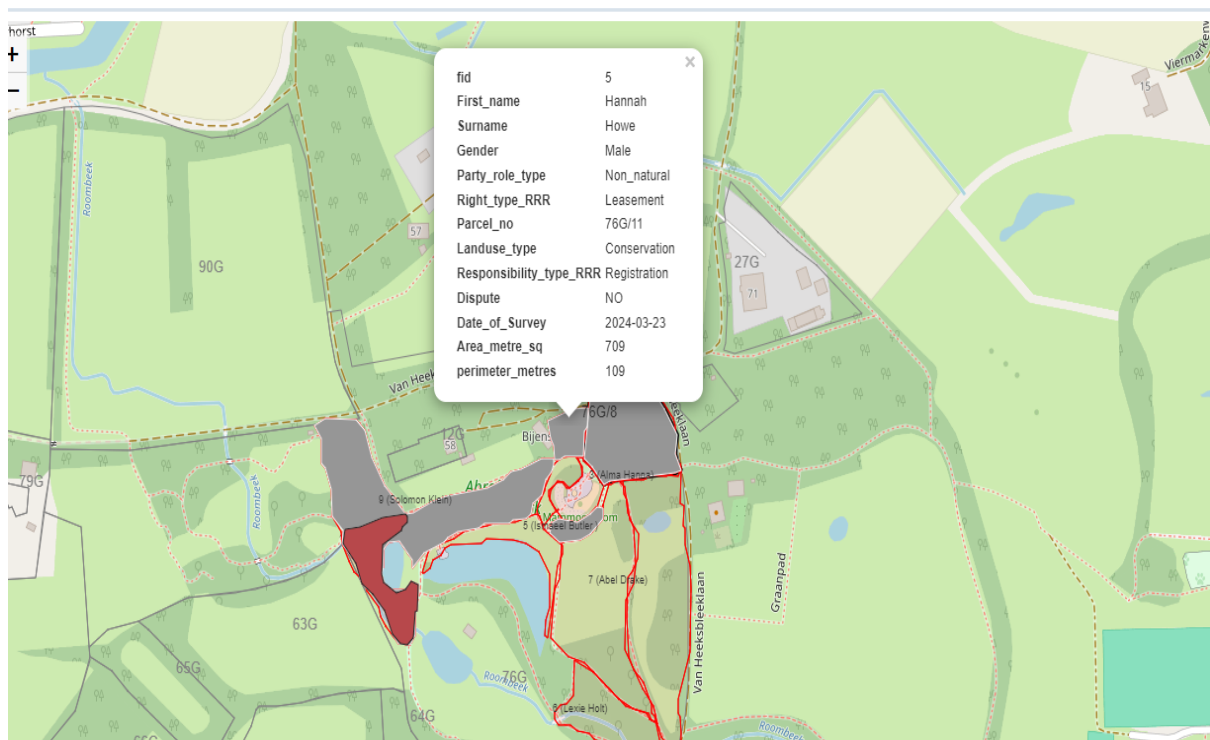


Figure 4 Qgis 2 web display

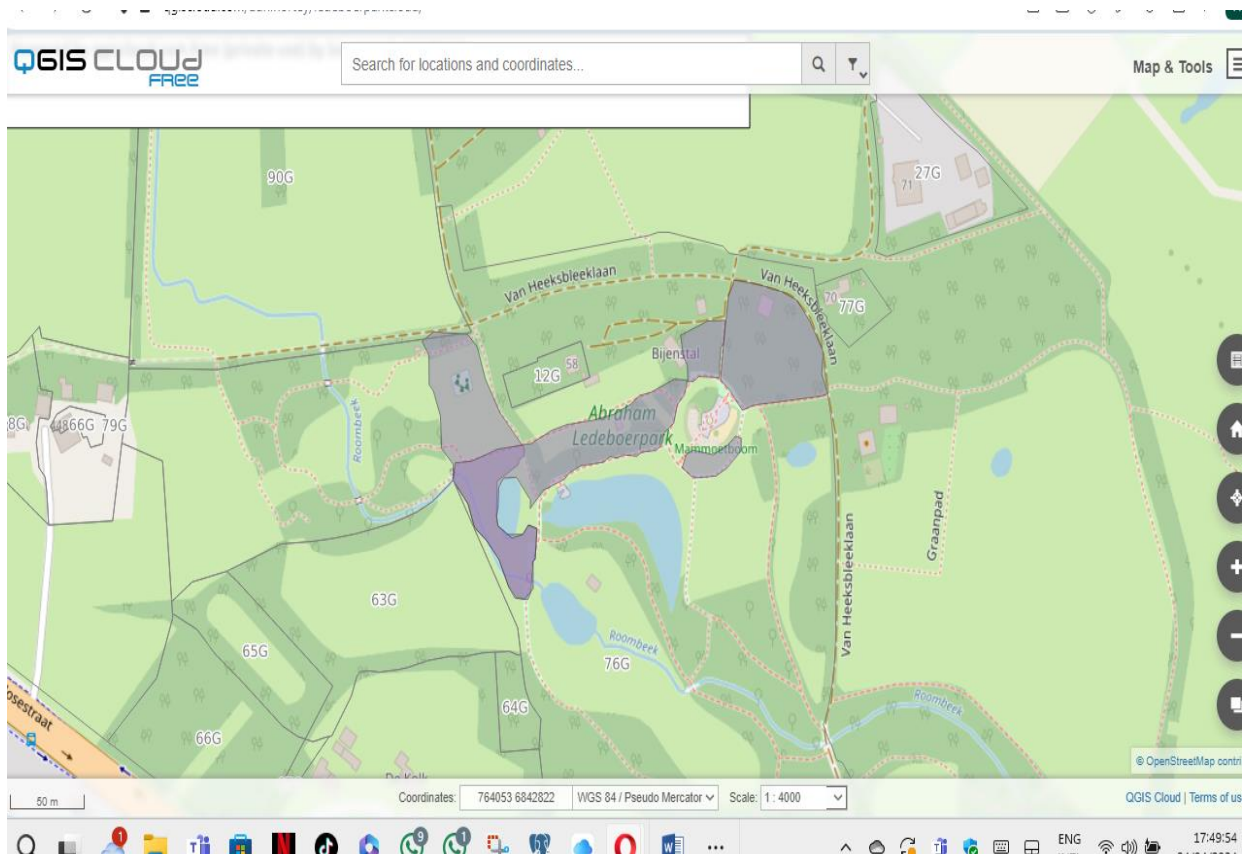


Figure 5 Qgis cloud display

6. Conflict documentation

In studies focused on renewing cadastres, it's crucial to identify overlapping parcels resulting from land measurements and document any conflicting areas. To streamline this process, the Geo package environment was chosen for its ability to store parcel geometry information in polygons. Utilizing the Intersect and overlay functions, overlapping polygons were identified to pinpoint conflicts.

The resulting overlapping areas, along with their corresponding attributes, are showcased in below and also illustrates the attribute table of the overlapping area, detailing the parties claiming the overlapping parcels, the rights they assert, the total area in overlap, and the percentage of the conflict area compared to each party's total claimed area.

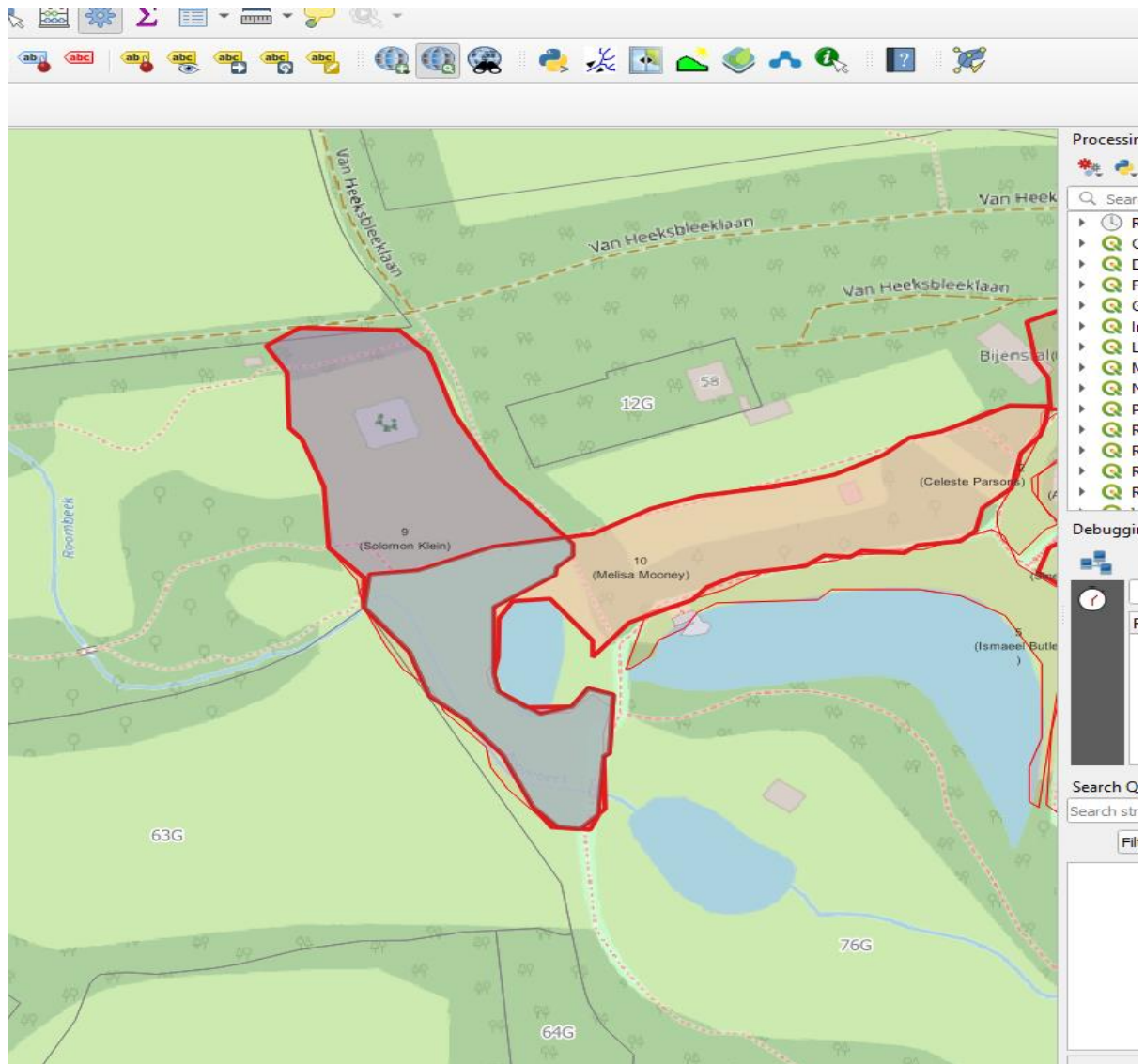


Figure 6 overlapping area of detected dispute

Party_ro_1	Right_ty_1	Parcel_no	Landuse_1	Restrict_1	Responsi_1	Date_of_1	Area	Peremeter
tural	Ownership	76G/10	Commercial	NULL	Registration	23/03/2023	1891.63225412000	260.67808704000

Figure 7 area of dispute in square m, needed to be addressed

7. Land title(s) for new parcels

In the final phase, the task of assigning titles to the measured parcels stored in the database remained. To achieve this, the QGIS print layout was used. The generated title comprised the party name, a unique spatial unit identifier, the associated right(s), and relevant cartographic elements.

As depicted in Figure 8, the land title also encompasses neighbouring parcels, providing a comprehensive overview. QGIS Print Layout offers an important feature known as Atlas, which streamlines the process of creating titles for parcels within the same layer. However, while the title in below benefited from the automatic table features of the QGIS print layout, the Atlas feature wasn't fully utilized. This was due to the title being generated from the title parcel table within the Post GIS environment, containing solely the parcel for which the title is intended.

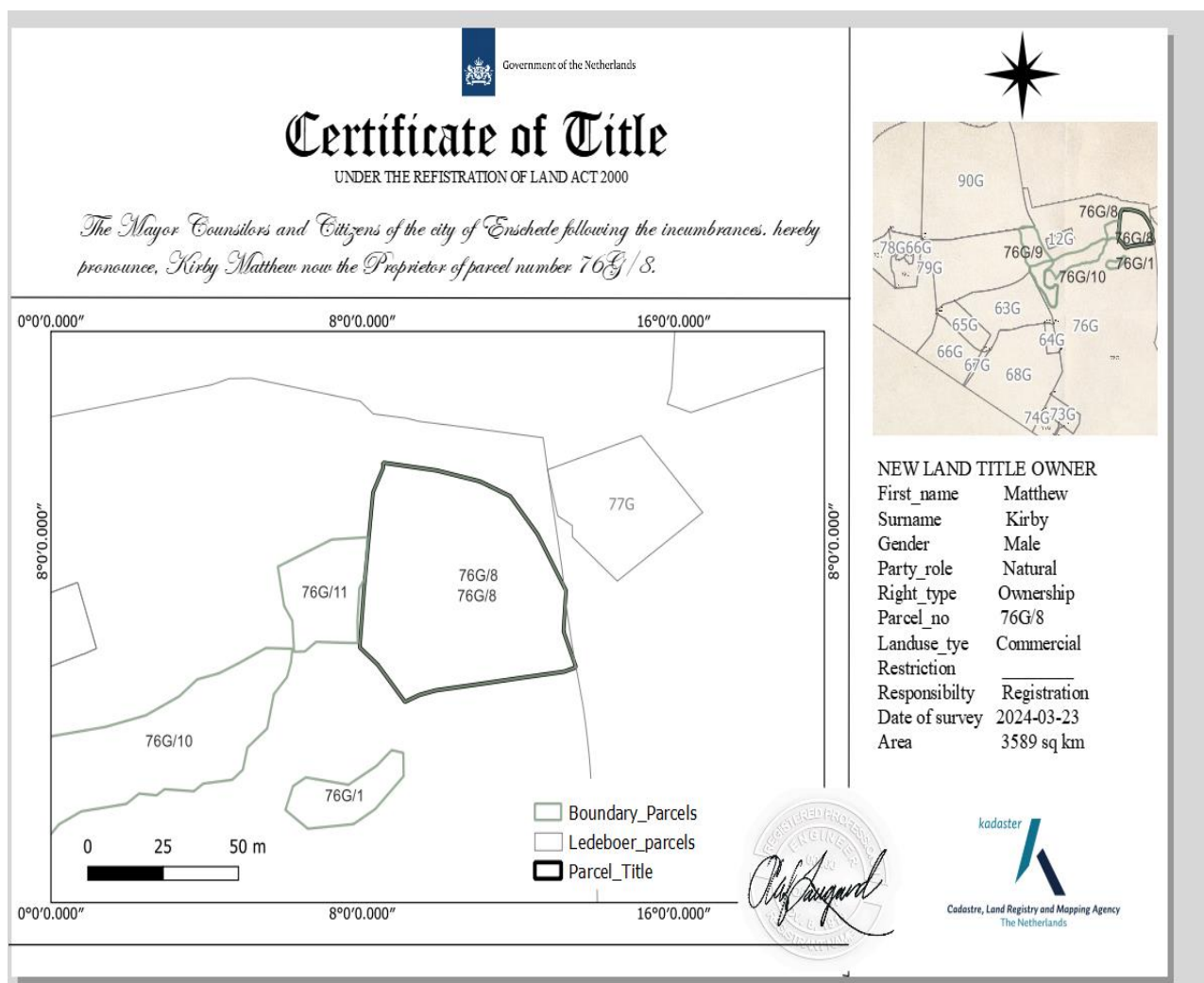


Figure 8 new land title for Matthew Kirby

8. Comparison tools

Stage/task	Suggested tool	Alternative tool	Tool used	Justification/rationale
Developing an LADM compliant data model	Enterprise Architect	Eclipse	Enterprise Architect	Enterprise Architect provides extensive UML modelling capabilities and robust support for the LADM data model. On the other hand, Eclipse may necessitate the addition of extra plugins to fully support the LADM data model
Implementing the LADM compliant data model	Geopackage file based database (QGIS)	Post GIS	geopackage	Geo package was easy to implement and deal with, as compared to the post GIS.
Field data collection	QField	Map it GIS	QField	QField stands out as a mature and widely adopted tool, offering strong support for offline data collection. In contrast, Map it GIS imposes limitations on the amount of data that can be measured, allowing only three data points in the free version.
Generating geo web services	QGIS Cloud	QGIS qgis2web	QGIS qgis2web and QGIS Cloud	QGIS2web offers greater flexibility compared to QGIS Cloud, as it accommodates multiple web mapping libraries and grants users greater autonomy in developing and deploying web maps. Meanwhile, QGIS Cloud facilitates the generation of a WMS link, allowing other users to access the contained file.
Web-map application for public inspection	QGIS cloud; Open Layers hosted on GitHub pages	QGIS qgis2web; leaflet	QGIS qgis2web; leaflet	Leaflet provides an extensive array of customization options and plugins, offering increased flexibility and control over the appearance and functionality of web maps and applications.

Issuing land titles	QGIS print layout,	ArcGIS print Layout	QGIS print layout	QGIS includes an Atlas feature, enabling the automated creation of multiple similar maps, which is particularly advantageous for issuing numerous land titles. Additionally, QGIS is freely available as open-source software.
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9. Advantages, and limitations

The system presents a blend of advantages and limitations that warrant careful consideration. While it effectively operates within confined spaces, implementing it nationwide may pose challenges. Despite facilitating various operations, the system incorporates manual calculations that could slow down procedures. Furthermore, the utilization of mobile devices for data acquisition yields both positive and negative outcomes. While it expedites data collection and reduces expenses, it may compromise data accuracy, leading to documentation conflicts.

QGIS Cloud, a data dissemination tool, proves effective in streamlining data publication through integration with the Leaflet Application. However, its performance may be inadequate for larger projects, where web services like GeoServer are recommended for handling larger datasets. Ensuring data backup is crucial, and cloud backup can be leveraged for this purpose.

QGIS Print Layout is a tool equipped with all necessary components to create titles, with the Atlas feature serving as an option to eliminate the need for additional requests to Post GIS. However, for this project, a separate table would have to be created in Post GIS for parcel titles containing only one parcel, thus bypassing the utilization of the Atlas feature.

The use of QField for field measurements has accelerated the workflow, but it carries the risk of inaccuracies that could result in significant financial and time losses for the organization. Although the current LADM tables provide a solid model for addressing complex cadastral problems, additional LADM tables may be required to enhance model clarity and trustworthiness when the project is implemented nationwide. The developed system significantly aids the municipality in updating land records. Leveraging the Land Administration Domain Model (LADM) facilitates the registration of land rights, while QField expedites the recording of missing cadastral data with minimal budget constraints.

Additionally, the use of PostgreSQL in data processing simplifies data management. However, if this system is to be applied nationwide, the accuracy obtained from mobile devices may prove inadequate, necessitating the use of GNSS measurement tools.

Furthermore, data dissemination may require the utilization of larger and more complex servers, such as GeoServer, capable of handling such data volumes. Lastly, data backup may be imperative to address potential vulnerabilities in the system.

10. Appendixes:

A-1: LADM Class diagrams in Enterprise Architect/sql codes

A-2: Geo package Database

A-3: Geo Web services Document

A-4: Layout of Land Title

URL (for HTML):

http://gisedu.itc.utwente.nl/student/s2781980/qgis2web_2024_04_26-14_05_24_745385/index.html#17/52.23470/6.86944

URL (Web map):

<https://qgiscloud.com/adhinortey/ledeboerparkcloud/>

Public (WFS/WMS/WCS):

<https://wms.qgiscloud.com/adhinortey/ledeboerparkcloud/>

